



Research paper

Real-time sonar simulation for UUVs using rasterization and omnidirectional screen space reflections

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ABSTRACT

Since real-world testing can be costly, risky, and difficult to reproduce, sonar simulation has become an important tool for developing and validating perception and navigation algorithms for unmanned underwater vehicles (UUVs). This work introduces a real-time sonar simulation framework integrated into the NauSim platform. Based on GPU rasterization and an omnidirectional extension of screen space reflections (SSR), the framework efficiently simulates primary echoes and approximates multipath effects. The proposed framework incorporates several acoustic and imaging phenomena, including Lambertian backscattering, signal attenuation, speckle noise, acoustic shadowing, and the influence of side lobes through beam-pattern modeling, while maintaining the computational performance required for real-time operation. In addition, dedicated anti-aliasing techniques, including procedural dithering and microtexture perturbations, reduce artifacts inherent to rasterization-based approaches. The framework is validated using real sonar data acquired with a Ping360 sensor mounted on a BlueROV2 platform. Quantitative and qualitative evaluations show good structural agreement between simulated and real sonar images. These results demonstrate that the framework provides a practical, computationally efficient tool for generating synthetic sonar datasets, thereby supporting the development and evaluation of autonomous underwater perception algorithms.

1. Introduction

Unmanned surface vehicles (USVs) and unmanned underwater vehicles (UUVs) have emerged as an important area of research and development for a variety of applications, ranging from environmental monitoring and infrastructure inspection to security and defense (Barrera et al., 2021). While USVs/UUVs offer significant advantages in terms of efficiency, safety, and coverage across these domains, developing robust control systems for marine navigation poses significant challenges. These stem from complex hydrodynamic forces, environmental disturbances (currents, wind, waves), and the requirement to perceive and react effectively to a constantly changing environment.

Consequently, simulators and digital twins have become essential for evaluating the operational integrity and performance of robotic systems. They enable the design and programming of control algorithms without the limitations imposed by physical hardware (Wibisono et al., 2023). Simulation not only mitigates the high costs and logistical risks associated with field experiments in hazardous or remote sites, but also allows for the validation of the vehicle's motion response in real time,

a critical capability for fully autonomous platforms with inherently unreliable acoustic links.

However, the effectiveness of these simulations in “in silico” environments depends largely on how well the environmental conditions are reproduced. The underwater environment presents unique perceptual challenges, characterized by high turbidity, backscattering, and rapid light attenuation in aphotic zones (McConnell et al., 2022). These conditions severely limit the range and resolution of conventional optical devices. Consequently, sonar (Sound Navigation and Ranging) systems have become one of the primary sensing modalities for underwater navigation and mapping (Zhou et al., 2023). Nevertheless, sonar data acquired on USV/UUV platforms are typically noisy, lower in resolution, and more difficult to interpret than optical imagery. This motivates the development of realistic and computationally efficient sonar sensor models that reduce the discrepancy between simulated and real-world observations. Such models facilitate the transfer of perception and navigation algorithms from simulation to underwater deployments while providing a controlled environment for their development and evaluation.

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Perhaps the most common approach to sonar simulation involves ray-tracing methods. These techniques simulate the propagation of sound waves through an environment by calculating intersections with 3D surfaces, explicitly computing successive acoustic paths through the scene, providing a direct approximation of the underlying physics. While these approaches offer high physical fidelity, their computational intensity has historically made them unfeasible for real-time applications.

However, these techniques have evolved significantly. Early foundational work, such as simple physics-based simulations (Linnett, 1997) and binary acoustic images (Gu et al., 2013), has given way to increasingly detailed approximations. For example, Kwak et al. (2015) extended ray-tracing to include continuous pixel intensities, while Mai et al. (2018) delved deeper into the physical modeling of sound interaction. More recently, platforms such as HoloOcean (Potokar et al., 2022) and newer simulators such as ACSim (Wang et al., 2025) have begun to expand their simulation models to address more complex phenomena associated with sonar imagery, extending the noise model or adding multiple reflections (multipath propagation).

On the other hand, since sonar simulation is a fundamental component of USV/UUV scenario modeling, its progress has mirrored the evolution of robotic simulation tools. Thus, taking advantage of the capabilities of the Gazebo robotic simulation environment (Nathan Koenig, 2004), various ray-tracing based sonar models have been developed. These typically utilize Gazebo's native ray-casting capabilities (DeMarco et al., 2015; Park, 2024; Choi, 2025), or repurpose GPU-optimized sensors originally designed for LiDAR simulation (Choi et al., 2021).

Concurrently, high-fidelity graphics engines like Unity3D and Unreal Engine have emerged as powerful alternatives. Their advanced rendering pipelines allow for more sophisticated environment modeling. Multiple works (Potokar et al., 2022; Almanza-Medina et al., 2019; Gentili et al., 2024) leverage the built-in ray-casting and shader architectures of these engines to emulate acoustic intensity through simplified physical models.

Advances in artificial neural networks (ANNs) have led to the creation of powerful tools for generating synthetic images. This has led to the emergence of work using neural networks (Sung et al., 2019b; Yu et al., 2023; Lian et al., 2025) or generative adversarial networks (GANs) (Sung et al., 2019a) to generate or improve sonar image quality. While these methods produce significant visual results, they only simulate the sonar image and do not consider any physical aspects of underwater acoustics.

Finally, it is worth mentioning rasterization-based simulations, as these were the first to achieve a level of performance fast enough for real-time simulations (Cerqueira et al., 2017, 2020). These methods are characterized by leveraging the graphics pipeline within a simulation engine to generate sonar images. This process incorporates previously calculated parameters such as pulse distance, return intensity, and field-of-view characteristics. While this concept trades physical acoustic realism for rendering efficiency, it represents an approximation between graphics and acoustics. Consequently, it tends to neglect real wave propagation phenomena, such as frequency-dependent attenuation, speckle noise patterns, or beam-pattern side lobes. Conversely, the flexibility and optimization of modern graphics pipelines make this approach a modular, efficient, and scalable solution with significant development potential.

This article presents an innovative real-time sonar simulation framework for unmanned underwater vehicles (UUVs), integrated into the NauSim platform (Ortiz-Toro et al., 2024), which optimizes the balance between computational efficiency and structural fidelity. Based on a revision of the pioneering work by Cerqueira et al. on GPU rasterization and focusing on mechanically scanning imaging sonars, the system features a novel approach to the simulation of secondary reflections, where acoustic propagation is reformulated as a reflection problem in image

space using a GPU; to do this, the system employs an omnidirectional extension of screen space reflections (SSR) based on 360° cubic maps.

Unlike ray-tracing approaches, the proposed omnidirectional SSR formulation estimates secondary reflections directly from rendered depth buffers. Although this approximation inherits the visibility limitations of screen-space techniques, it eliminates repeated scene-intersection queries and the associated scene-based spatial hierarchy optimizations while preserving the dominant multipath effects relevant to perception-oriented sonar simulation. As a result, the proposed method models complex multipath interactions and secondary reflections at only a fraction of the computational cost of conventional ray-tracing methods.

Beyond secondary reflections, the framework incorporates several key acoustic phenomena, including but not limited to attenuation, speckle noise, advanced backscattering models, reduction of rendering artifacts using dithering techniques, and the influence of side lobes on the image. To the best of our knowledge, this is the first sonar simulation framework to integrate all these components within a unified, real-time GPU-based pipeline. The proposed approach is evaluated both quantitatively and qualitatively using real sonar data acquired with a Ping360 mounted on a BlueROV2 platform. A comparative summary between the state-of-the-art works and the presented method is detailed in Table 1.

The objective of this work is not to solve the complete underwater acoustic propagation problem, but to generate computationally efficient sonar images that reproduce the structural characteristics and major acoustic phenomena required for perception, navigation, and algorithm development.

The remainder of the paper is organized as follows: Section 2 provides an overview of the foundational concepts, including sonar devices, the GPU general pipeline, and SSR. Section 3 details the materials and methodology employed. The experimental results are presented and discussed in Section 4, while Section 5 offers concluding remarks and explores potential future work.

2. Background

The following section summarizes the key concepts underlying the proposed framework, including sonar imaging principles, the GPU rasterization pipeline, and the basis of the SSR methods used for multipath simulation.

2.1. Sonar devices

Sonar systems are active or passive devices that use the propagation of sound waves to detect and locate underwater objects. While passive sonar is limited to perceiving ambient sounds, active sonar emits acoustic pulses and analyzes their echoes. When the acoustic signal emitted by an active sonar encounters a surface, part of the signal is absorbed and another part is reflected. The analysis of the reflected signal is the basis for determining an object's distance through time-of-flight and its nature based on the intensity of the return (Urick, 1983).

In the context of a sonar system, the visible or insonified area is delimited by a specific aperture, determined by the azimuth (θ) and elevation (ϕ) angles and the signal's reach or range (r). Each reading in a specific direction forms a "beam", and each distance sample within that beam is called a "bin". During the insonification process, the elevation angle, denoted as ϕ , is lost. Consequently, a sonar system provides a range r at an azimuth angle, resulting in a projection of three-dimensional information onto a two-dimensional plane. Thus, the information is defined in a polar coordinate system (Johannsson et al., 2010). It is important to note that data quality can be affected by various factors, such as acoustic shadows (areas behind objects that block sound), reverberation from multiple echoes, and interference or speckle noise. Furthermore, these distortions can intensify during the

Table 1

Comparison of representative sonar simulation frameworks, including the proposed approach, in terms of the underlying simulation technique, real-time performance (at the time of the framework's publication), modeled acoustic effects (multipath, attenuation, noise, and side lobes), reflection model, and validation strategy.

Reference	Simulation	Real time	Multi-path	Attenuation	Noise	Side lobes	Reflection model	Validation
Almanza-Medina et al. (2019)	Ray tracing	No	No	Range falloff	Gaussian & Rayleigh	No	Lambert	Quantitative
Cerqueira et al. (2020)	Rasterization & Ray tracing	Yes	Yes	Spreading & medium absorption	Multiplicative	No	Lambert	Qualitative, Quantitative, Performance
Potokar et al. (2022)	Ray tracing	Yes	No	Range falloff	Additive & multiplicative	No	Lambert	Qualitative
Yu et al. (2023)	Generative Adversarial Network	Yes	Implicit	Implicit	Implicit	Implicit	Learned	Qualitative, Quantitative
Gentili et al. (2024)	Ray tracing	Yes	No	Acoustic absorption	Multiplicative	No	Lambert & material properties	Qualitative
Choi (2025)	Ray-Based point scattering model	No	No	Spreading & acoustic absorption	Beam interference	No	Point scattering model	Qualitative, Quantitative
Wang et al. (2025)	Ray tracing	No	Yes	Physical acoustic propagation	Rayleigh & Multibeam artifacts	Yes	Bidirectional reflection distribution	Qualitative, Quantitative
Proposed	Rasterization & Omnidirectional SSR	Yes	Yes	Medium absorption	Multiplicative & microtexturing	Yes	Lambert/Cosine-squared/Oren-Nayar	Qualitative, Quantitative, Performance

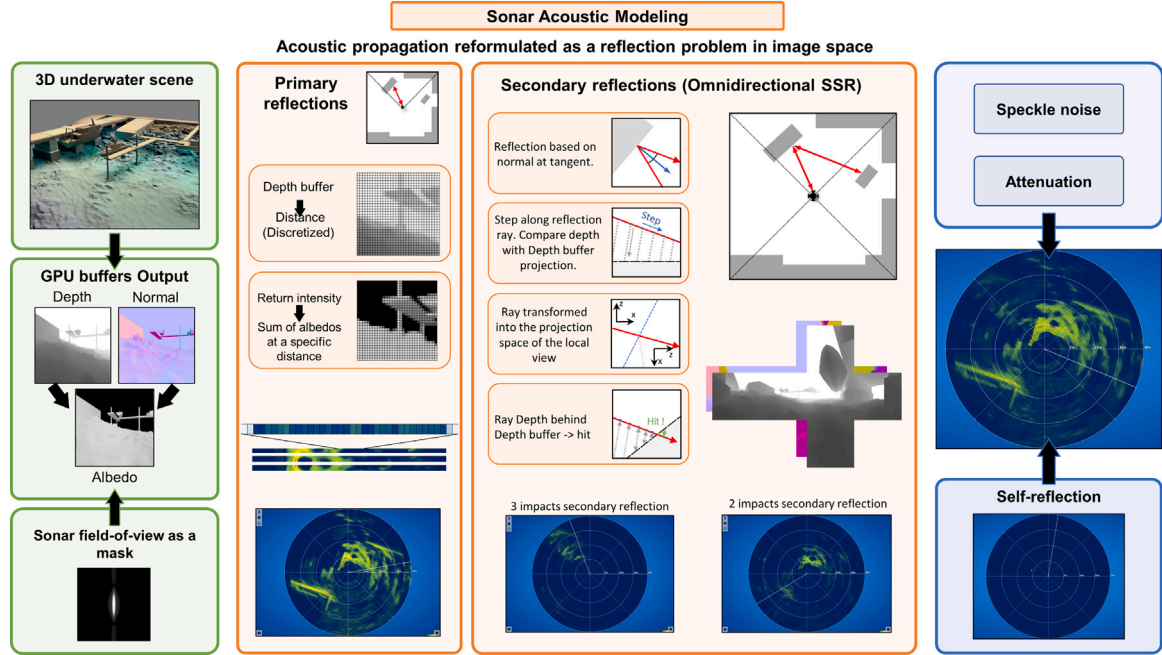


Fig. 1. Diagram illustrating the rasterization-based sonar simulation process. The system reformulates acoustic propagation as a reflection problem in image space and uses the GPU's depth and normal buffers to compute primary echoes. For secondary multipath interactions, it employs an omnidirectional reflection approach in screen space. Post-processing stages incorporate physical phenomena, such as speckle noise, signal attenuation, and self-reflection.

conversion of the original data from polar to Cartesian coordinates, which is performed to facilitate interpretation.

The design of any sonar system involves a critical trade-off between range and resolution, dictated by the operating frequency. Low frequencies achieve greater propagation distances but offer limited detail. Conversely, high frequencies provide superior resolution, although their operating range is considerably restricted. Given the nature of these vehicles, it is common for high-frequency devices to be mounted on USVs/UUVs (Cox, 2004).

2.2. GPU rendering pipeline and screen space reflections

The GPU rendering pipeline is the sequence of programmable and fixed-function stages that transforms a three-dimensional scene into a

final, two-dimensional image. This highly parallel process begins on the CPU, where the rendering engine prepares geometry, materials, lighting, and camera data before sending them to the GPU through draw calls. Once transferred to GPU memory, rendering is divided into three main stages: geometry processing, rasterization and fragment processing, and post-processing. During geometry processing, vertices are transformed into screen space and assembled into primitives for rasterization. Fragment processing then computes the appearance of each visible pixel by evaluating materials, textures, and lighting models, storing the resulting information in several frame buffers. These buffers, which include color, depth, normals, and various material properties, are available for processes such as lighting calculations and post-production tasks like tone mapping, edge smoothing, the "bloom" effect and motion blur. The intermediate data generated

during these processes also forms the basis for various screen-space rendering techniques, such as screen space reflections (SSR).

Screen space reflections (Sousa et al., 2011; McGuire and Mara, 2014) is a post-processing technique that generates dynamic reflections by reusing the information already stored in the depth and color buffers, rather than tracing light rays throughout the entire three-dimensional scene. This approach considerably reduces the computational cost compared with ray tracing while still producing visually convincing reflections. SSR commonly relies on a ray-marching algorithm, where, for each reflective pixel, a reflection ray is computed from the viewing direction and the surface normal and advanced through screen-space coordinates in discrete steps. At each step, the ray depth is compared with the depth stored in the depth buffer. When an intersection with the recorded scene geometry is detected, the corresponding color from the color buffer is sampled and used as the reflected contribution. As SSR operates exclusively on the information available in the current frame, it is inherently limited to the camera's field of view. Objects that are off-screen or occluded by other geometry are absent from the G-buffer and therefore cannot appear in the reflections, often producing missing regions or other visual artifacts. Nevertheless, SSR remains one of the most widely used reflection techniques in real-time rendering because it offers a favorable compromise between visual quality and computational efficiency.

3. Materials and methods

In underwater research exploration, before testing on a real platform, a design, validation, and testing process is carried out in a simulated environment. It is in this phase, in a simulated environment, that the need arises for a realistic sonar signal that provides the virtual vehicle with sensorization equivalent to that which will be used in a real environment.

3.1. Sonar rendering simulation

The implementation of sonar simulation (see Fig. 1) is proposed as an extension of a general GPU rendering pipeline, conceived as a deferred path. In this rendering model, a separation is made between geometric rendering and lighting. In a first pass, object data (color, normals, depth) is recorded in memory buffers. Subsequently, this data is used in a second pass to calculate the lighting of the entire scene simultaneously. These buffers are used as a basis for generating the acoustic reflection model.

The insonification model requires three parameters for its correct implementation: pulse distance, echo intensity, and sonar field of view. The pulse distance represents the estimated time-of-flight of the acoustic signal. This estimate is derived from the depth buffer by calculating the distance between the sonar emission point and the visible surfaces. The sonar field of view reflects the specific aperture characteristics of different sonar models. Depending on the type, this may involve dividing the image into multiple beams, each of which is independent. The rendered image is subdivided into beams according to the angular position.

For an ideal diffuse reflector, the intensity of backscatter will be uniform in all directions from the reflecting surface. In other words, when modifying the orientation of the receiving transducer while keeping the position of the sound source constant, the energy reflected at a specific solid angle from the surface is identical in all directions. However, a variation in intensity depending on the direction of insonification is observed. While most marine materials are not perfect diffuse reflectors, Lambert's law provides a reliable approximation for acoustic backscattering in aquatic environments (Jung et al., 2018). Therefore, the intensity is analyzed by examining the normals of the objects present in the simulation and the direction of sonar emission, taking Sac et al. (2015) as a reference.

The implementation encodes the rasterization data into 8-bit RGBA channels. In the simulator, the RGB channels store information about the echo, the pulse distance, and the sonar field of view. The echo intensity is represented by 8-bit values; the pulse distance is managed using GLSL's 32-bit depth buffer to avoid precision issues. The three parameters are normalized between 0 and 1. Intensity ranges from absence to maximum energy, distance from near to far objects (limited by the sonar's range), and the field of view is defined with the zero angle at the center of the image, increasing on both sides to the maximum allowed by the sonar. The image is divided into beams according to the horizontal field of view, assigning each column an angle and a beam vector. Each beam has pixels grouped into range cells using a histogram based on pulse distance and echo intensity. The number of range cells is determined by the number of samples configured in the target sonar to be simulated. These containers represent increasing ranges, from the closest to the farthest objects, using a distance histogram that accumulates the corresponding intensity. Each of the simulated phenomena (i.e., primary echoes, secondary echoes, self-reflection) adds the intensities of its echoes in each bin.

Due to water absorption by the acoustic wave, the echoes from the final bins are weaker than those from the first ones, as the energy dissipates into the environment. The transmission loss suffered by the acoustic signal in salt water is modeled based on Thorp (1967), simplified as the attenuation coefficient α (in dB/km):

$$\alpha = \frac{0.1f^2}{1+f^2} + \frac{40f^2}{4100+f^2} + 0.000275f^2 + 0.003$$

where the sonar frequency f is in kHz. The first term (~ 1 kHz) represents the relaxation of boric acid (H_3BO_3), the second term (~ 65 kHz) represents the relaxation of magnesium sulfate (MgSO_4), and the third term represents viscous absorption in fresh water. Depending on the scenario to be simulated, either the complete model or only the term corresponding to fresh water will be used.

Since the sonar image is calculated at the pixel (texel) level, it is possible to assign characteristics to each element of the simulated scene so that they are interpreted as part of the sonar simulation using textures, such as reflectivity coefficients for echo intensity modulation or small-scale details using normal mapping.

Finally, to increase the realism of the sonar image, multiplicative noise, specifically speckle noise (Jaybhay and Shastri, 2015), has been implemented. The phenomenon of speckle noise is an intrinsic and recurring feature in sonar images, derived from the inherently coherent nature of sonar signals. This phenomenon originates as a result of the constructive and destructive interference of sound waves backscattered by small surface features or particles in the aquatic environment.

3.2. Primary reflections

In sonar systems, primary reflection refers to the direct, specular echo from the initial surface of a target object. This element represents the fundamental return of the "ping", enabling the system to calculate range. Consequently, other phenomena occurring in conjunction with a sonar ping are often categorized as noise or artifacts.

Primary reflections are derived from a primary camera within the simulation. This camera is configured to emulate the specific characteristics of the sonar model being simulated. In some cases, it may be necessary to apply a mask to ignore specific portions of the primary camera's image.

The process is implemented as a post-processing element, specifically a compute shader. It functions by extracting data from the depth and normal buffers, allowing the program to identify surface orientations and their distances from the sensor. The depth buffer is then normalized; based on this data, the coordinates of each pixel are reconstructed as points within the camera space.

Once the depth map has been obtained and normalized, the shader approximates the physical behavior of sound using Lambertian reflection calculations. At this stage, the intensity of the sound bouncing

back to the sensor is determined. Surfaces oriented directly toward the camera generate a stronger echo, while inclined or distant surfaces produce a weaker signal. This reflection factor is then used to visually represent the intensity of the acoustic return within the output textures.

The final step is the generation of the histogram, which constitutes the sonar signal itself. The shader calculates the “range” (distance) of each point from the depth buffer and accumulates these results in the output buffer. It should be noted that the geometry of the latter will depend on the specific characteristics of the sonar return, such as the number of samples per return and the system’s beam count. The final result is a graphical representation of energy, where the distance axis displays intensity peaks in areas containing solid objects. This representation closely mimics the data a physical sonar sensor would provide in an underwater environment.

3.3. Secondary reflections

The multipath phenomenon represents some of the most noticeable “artifacts” in a sonar signal. It is, therefore, one of the main differences between an ideal simulated image and a real sonar image. This phenomenon originates from sonar signals that follow multiple paths between a transmitter and a receiver. The creation of multiple signal paths occurs when a sonar signal is reflected at one or more interfaces, and the reflected signals combine with the direct path signal, causing constructive and destructive interference. The signal received by the sonar may present multiple echoes for the same structure, coming from the same direction or from another, which complicates the interpretation of the resulting signal.

Ray tracing is the most common approach to simulating secondary reflections in sonar, particularly in GPU-based simulators that use rasterization. By tracing successive acoustic paths explicitly, recursive ray-tracing methods can accurately model multipath propagation and higher-order reflections, providing high physical and geometric fidelity. However, this level of accuracy incurs a significant computational cost, as performance is dependent upon overall scene complexity, recursion depth, and the specific acceleration structures implemented in the 3D engine to optimize the repeated scene intersection queries (Xu et al., 2024).

In contrast, the proposed approach reformulates secondary acoustic propagation as a screen space reflection problem. Instead of launching additional rays into the scene, secondary echoes are estimated through omnidirectional screen-space ray marching, which is performed directly on the depth and geometric information generated during the primary rendering pass. Consequently, the method reuses existing frame buffers without requiring a dedicated ray-tracing architecture or additional scene representations. As a result, it seamlessly integrates into the rasterization pipeline and is, to an extent, independent of the rendering engine.

The computational cost of this method is primarily determined by image-space parameters such as framebuffer resolution, ray-marching length, and the number of processed pixels (Beug, 2020). This makes execution time more predictable and largely independent of the geometric complexity of the scene. While it shares the visibility limitations of screen-space techniques, it effectively reproduces the dominant multipath structures necessary for perception-oriented sonar simulation, all the while maintaining real-time performance.

Although SSR can be used to calculate secondary reflections, the typical formulation of this technique is limited by the field of view of the main camera, whereas secondary reflections can occur from any object surrounding the sonar. In this case, a formulation that extends classical SSR to a fully omnidirectional representation is proposed, allowing for the reconstruction of seamless 360° reflections while maintaining the computational efficiency of depth-based ray marching. The environment is represented by a cube map composed of six perspective cameras aligned with the cartesian axes $\{\pm X, \pm Y, \pm Z\}$ and centered on the vehicle. Each face stores normal and depth maps. Given a surface

point $\mathbf{P} \in \mathbb{R}^3$, reconstructed from the depth and its normal $\mathbf{N}$, the reflected signal is estimated by tracing a reflection ray within the domain of the cube map and checking for intersections with the stored depth values. The center of the cube map is defined as the point $\mathbf{C}$, where the sonar emitter is placed in virtual space. The view direction is defined as:

$$\mathbf{V} = \frac{\mathbf{P} - \mathbf{C}}{\|\mathbf{P} - \mathbf{C}\|}, \quad (1)$$

and the reflection direction, for each pixel in the sonar field of view is computed using the standard reflection equation.

$$\mathbf{R} = \mathbf{V} - 2(\mathbf{V} \cdot \mathbf{N})\mathbf{N}. \quad (2)$$

The reflection ray is parametrized as

$$\mathbf{X}(t) = \mathbf{P} + t\mathbf{R}, \quad t > 0. \quad (3)$$

The core of the algorithm is a ray-marching loop. In each iteration of the ray-marching process, the algorithm moves the ray along the reflection direction and verifies the correspondence between the direction from the sonar position and the cube’s faces. In this way, it determines whether the ray has intersected the geometry. If the ray reaches the edge of one of the faces, it is transferred to the adjacent face, maintaining continuity along the edges of the cube. Once the face is identified, the current position of the ray is transformed into the projection space of that face’s local view, and the corresponding depth map is sampled to check for intersection. The face is chosen according to:

$$\text{face} = \begin{cases} +X & |x| = m, x > 0, \\ -X & |x| = m, x < 0, \\ +Y & |y| = m, y > 0, \\ -Y & |y| = m, y < 0, \\ +Z & |z| = m, z > 0, \\ -Z & |z| = m, z < 0. \end{cases} \quad (4)$$

Thus, transitions between cameras occur implicitly as the dominant coordinate changes during ray propagation.

The selected 3D position is projected onto the corresponding 2D face coordinates $\{\pm u, \pm v\}$. For example, for the $+X$ face,

$$u = \frac{-z}{|x|}, \quad v = \frac{-y}{|x|}, \quad (5)$$

with analogous permutations for the remaining faces. At each iteration, the ray depth is compared to the stored depth value,

$$\Delta = z_{\text{ray}} - z_{\text{depth}}(u, v), \quad (6)$$

and an intersection is detected when

$$|\Delta| < \epsilon \quad \text{or} \quad z_{\text{ray}} > z_{\text{depth}}. \quad (7)$$

The step size is reduced when a depth sign change is detected to improve intersection accuracy without resorting to full binary search.

Once an intersection is detected, the process is similar to that followed in primary reflections. The Lambertian reflection of the reflected ray is calculated from the normal of the cubemap at position (u, v) , as well as the determination of the distance from the primary reflection point to the intersection. The intensity of the sound transmitted to the device by a secondary reflection is determined by the sum of the impacts experienced by the ray on its path. At present, two cases are being considered (see Fig. 2): three impacts, where the intensity is defined by the initial impact, the reflected impact, and the initial impact on the return path; and two impacts, where the distance to the sonar is equivalent to the path formed by the primary impact, the secondary impact, and a direct path to the sonar. Due to the geometry of the sonar receiver, two impacts should only be detected when the return signal is received from the front. In other cases, the signal should be either absent or significantly reduced.

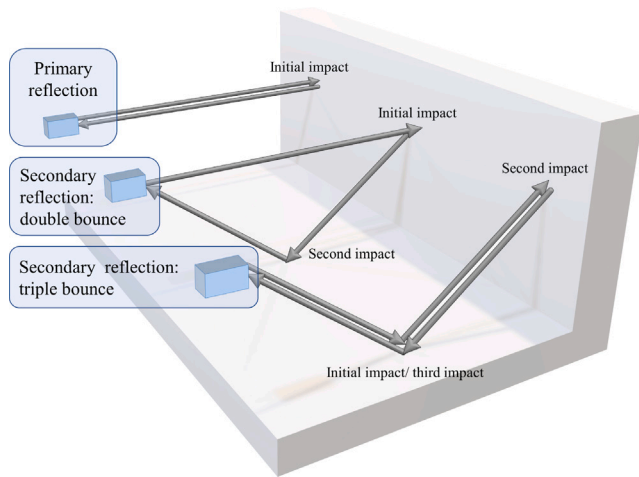


Fig. 2. Schematic representation of simulated acoustic ray paths. The diagram shows direct primary reflections and multipath phenomena: secondary reflections resulting from double and triple bounces off the surrounding geometry.

3.4. Insonification method

As mentioned previously, Lambert's law, defined as $I = \cos(\theta)$, where θ is the angle between the surface normal vector $\vec{N}$ and the incident acoustic wave direction $\vec{L}$, provides a simple yet effective approximation of acoustic backscattering and has been adopted in several sonar simulation approaches. Lambert's cosine law assumes an ideal diffuse reflector, scattering the incident energy uniformly across all outgoing directions regardless of the viewing angle.

However, the classical Lambert model is not the only formulation reported in the literature. A commonly used empirical variation replaces the cosine term with a squared cosine, $\cos^2(\theta)$, producing a steeper angular falloff and a more pronounced directional response. This modification is useful when the surface is expected to exhibit stronger specular characteristics or a narrower scattering lobe than predicted by the standard Lambertian model. Both Jackson et al. (1996) and Thorsos et al. (2001) consider the energy received from insonification to be better modeled as the squared cosine.

A more physically motivated extension is the Oren–Nayar reflectance model (Oren and Nayar, 1994), which was adopted by Kun et al. (2021). This model recognizes that perfectly Lambertian surfaces rarely exist in nature (Kienle and Foschum, 2011). Instead of modeling the surface as an ideal smooth, flat plane, the model represents it as an ensemble of microscopic V-shaped cavities that account for surface roughness. The backscattered intensity is influenced by both the incidence angle and the observation (or viewing) angle relative to the surface. When the observation direction is close to the illumination direction, the micro-cavities exhibit increased inter-reflection, reducing the edge darkening present in Lambertian models. As a result, the Oren–Nayar model provides a closer approximation of the scattering from rough, diffuse surfaces.

Given the importance of the insonification method in a sonar simulation, tests will be carried out using the three methods mentioned above. Fig. 3 shows a graphical representation of the acoustic dispersion methods being tested.

3.5. Sonar image “artifacts”

Although the phenomenon of multipath propagation is one of the most obvious “artifacts” in the Sonar image, it is influenced by various factors stemming from the specific properties of the simulated hardware and the configuration of the simulation scenes.

3.5.1. Secondary lobes simulation

The term “side lobes” in sonar technology refers to unwanted beams of acoustic energy that propagate beyond the main beam axis due to diffraction at the edges of the transducer. In general, a well-designed single-beam transducer will exhibit side lobes with sensitivity far lower than that of the main lobe, but completely eliminating side lobes is virtually unattainable. The presence of side lobes results in unwanted side echoes that generate clutter and “wing-shaped” artifacts that can obscure real targets or corrupt data, especially near the surface or the bottom (Blondel, 2010).

The amplitude of the sonar signal geometry can be modeled using Fraunhofer's diffraction principle, which states that the radiation pattern in the far field is the spatial Fourier transform of the amplitude distribution (physical aperture) of the transducer (Urick, 1983). Thus, the amplitude distribution of the sonar signal acts as a signal in the spatial domain, and the beam pattern (including the main beam and side lobes) is its representation in the spatial frequency domain. The goal is to visualize how sound energy is distributed in the plane perpendicular to the direction of propagation, which will give us a mask representing the front geometry (the radiation pattern) of a sonar beam. Thus, instead of simply using a mask to limit the field of view, we can insonify the scene based on a representation of the beam geometry, with specific weights for each of its pixels. An 8-bit grayscale mask allows us, theoretically, to represent an attenuation of up to -24 dB, or 0.39% ($1/255$) of the main signal.

3.5.2. Aliasing

In GPU-based sonar simulation, the banding phenomenon occurs because submerged surfaces or the air–water interface may lie nearly parallel to the sonar beam. This situation renders the precision of the depth buffer (typically 16 or 24 bits) insufficient. At these grazing incidence angles, the discrete jump between consecutive depth values is greater than the actual physical slope of the terrain. This limitation in grid discretization prevents the capture of subtle, continuous changes in distance, resulting in a “staircase effect”. As a result, the simulator assigns the same angle of incidence to large groups of pixels, appearing visually as concentric rings of uniform brightness, rather than a smooth, realistic backscatter gradient.

To address this issue, two complementary approaches are implemented, each tailored to the nature of the reflection. To perform primary reflections, a procedural dithering technique is implemented that adds a slight, random perturbation to the depth map. Since this process has the potential to alter or obscure certain details, this stochastic noise is modulated so that it is applied exclusively in areas where banding may occur, that is, areas where the view is more oblique. These areas are determined by evaluating the relationship between the normal and the view vector. This perturbation effectively hides the bands resulting from discretization by adding small variations to the angle of incidence in groups of adjacent pixels.

On the other hand, for secondary reflections, a more diegetic approach was chosen by applying microtextures to the surface normals. This technique works similarly to dithering, as it breaks the uniformity of the reflections and mitigates the banding effect, since each pixel within what was previously a ring of uniform brightness now has a slightly different normal ($\mathbf{n} + \Delta\mathbf{n}$). Additionally, it adds a layer of physical realism; by representing small structures and irregularities in the seabed and marine structures, it approximates the acoustic response of small-scale roughness by effectively simulating the noise associated with secondary reflections.

3.5.3. Vehicle self reflection

A common form of secondary reflection is what is known as secondary self-reflection. After bouncing off an external object, the sound returns to the underwater vehicle's hull, bounces off it again, and reflects back to the object before finally returning to the transducer. This phenomenon occurs most frequently in confined or highly reflective

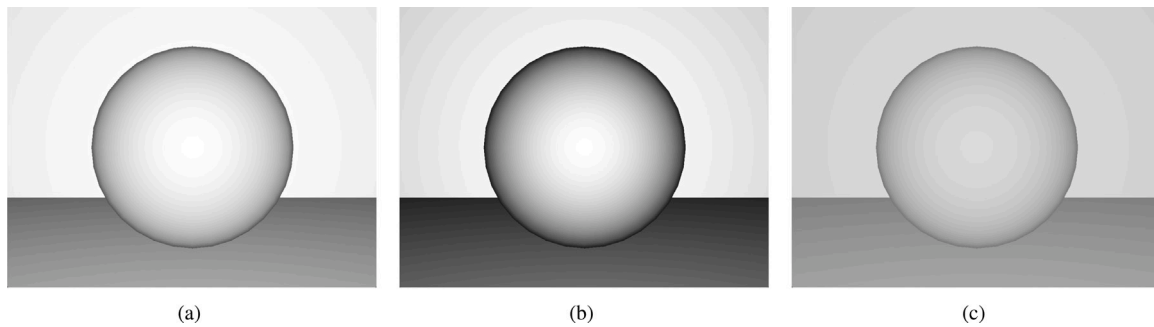


Fig. 3. Visual comparison of the acoustic backscatter models evaluated in the simulation. Representation of the acoustic return from a sphere: (a) corresponds to the classic Lambert's cosine law. (b) shows the squared-cosine approximation. (c) shows the Oren–Nayar model.

environments, such as the interior of offshore structures, ports with concrete walls, or when the vehicle operates in close proximity to the seabed. Given that the camera configuration geometry used for calculating reflections with SSR does not allow visibility of the vehicle, it is not possible to perform secondary self-reflections directly.

Since, in the test scenarios, this phenomenon appears as a series of small but highly characteristic echoes, it was decided to implement a simplified approach. Thus, the presence of the vehicle on the screen is approximated as a small sphere located at the center, thereby verifying whether a collision with that object occurs during the ray-casting process.

4. Results

The simulation of a Ping360¹ is proposed as the target for testing. The Ping360 is a Mechanical Scanning Imaging Sonar (MSIS), designed for use in ROVs or other underwater vehicles. An MSIS uses a single transducer (transmitter/receiver) mounted on a stepper motor. The sensor emits an acoustic pulse in a specific direction, listens for the echo, and then rotates a small angular increment before emitting the next pulse. The Ping360 has an angular increment of 0.9° , its maximum range is 50 m, and has an operating depth of 300 m. It operates at an approximate frequency of 750 kHz, offering a 25° vertical and 2° horizontal beamwidth. The minimum detection range is approximately 0.75 m, with a relative resolution of around 0.08% of the range.

The Ping360 is mounted on a BlueROV2 UUV, in its heavy configuration (Robotics, 2016). The simulation has been programmed using GLSL and is fully integrated into NauSim, which was developed in Python. This environment manages the interactions and physics associated with the virtual vehicle and allows direct control of the real vehicle(s) for the implementation of the control models developed.

The synthesized sonar data is encapsulated within UDP ping-protocol frames.² In this way, it is possible to use the same application as the original hardware, pingViewer™, to visualize the data in real time.

The primary camera in the simulation is configured using a 1024×1024 pixel buffer. Once the corresponding masks have been applied to approximate the sonar's field of view, 91,548 pixels remain for the primary echo simulation. The secondary echo simulation is more computationally expensive; therefore, a smaller buffer of 512×512 is used for each of the cameras that form the cubemap. Once the corresponding masks have been applied to the secondary frontal camera, the secondary echoes are simulated by performing ray marching from 24,184 pixels.

Once the buffer resolutions have been defined, the behavior of the proposed SSR algorithm is governed by three main parameters:

the marching tolerance ϵ , the maximum number of iterations, and the initial step size. The maximum number of iterations establishes the stopping criterion when no intersection is found and therefore directly influences computational performance. The marching tolerance, ϵ , determines the depth precision required to declare a ray-buffer intersection. Larger values reduce intersection accuracy and increase the likelihood of aliasing artifacts, especially at grazing angles and in the connections between faces on the reflection cube. Smaller values improve precision but degrade performance. The initial step size, expressed as a fraction of the normalized reflected vector, controls the distance advanced by the ray during each marching iteration. Larger step sizes accelerate traversal, but may miss portions of the scene, producing ring-shaped artifacts. Conversely, smaller step sizes improve the detection of fine details, but reduce the effective propagation range of the ray for a fixed iteration amount. Fig. 4 illustrates the influence of these parameters in a simple test environment, showing both the image reflection maps and the corresponding simulated sonar images.

The values of these parameters were selected empirically through extensive experimentation. Alternative configurations either produced negligible improvements in computational cost or degraded it, or introduced visible rendering artifacts while increasing.

A summary of the Ping360 specifications and the SSR configuration parameters used in the experiments can be seen in Table 2.

Tests are conducted in two different scenarios. The first (Scenario A) corresponds to the pool at the Ports Laboratory of the UPM School of Civil Engineering, a $33 \times 11 \times 1.36$ meter pool equipped with a wave generator used to conduct hydrodynamic tests on marine structures. Based on this scenario, three scenes are proposed.

- Scene 1: The sonar is set to a range of 15 m, with the vehicle 11.8 m from the wave generator, at a depth of 0.60 m. Within the range of vision, some modeled port structures can be observed.
- Scene 2: The sonar is set to a range of 15 m; the vehicle is 6.85 m from the wave generator and at a depth of 0.8 m.
- Scene 3: The sonar is set to a range of 10 m; the vehicle is 5.5 m from the wave generator, 1.2 m from the edge of the pool, and at a depth of 1 m. There is an obstacle (another BlueROV2) located approximately 1.1 m from the transmitter.

The second scenario (Scenario B) corresponds to the Hydrodynamic Testing Channel (CEHINAV) at the UPM's Higher Technical School of Naval Engineering. This is a $100 \times 3.8 \times 2.2$ m channel equipped with a wave generator, which is used in hydrodynamic studies to optimize the shapes of ships and other marine structures. Based on this scenario, two scenes are proposed.

- Scene 1: The sonar is set to a range of 5 m, with the vehicle at a depth of 0.6 m, 1.2 m from the edge of the channel, and 2.2 m from the wave generator.
- Scene 2: The sonar is set to a range of 5 m. The vehicle is positioned at a depth of 1 m, 2.7 m from the wave generator structure, and at a distance of between 0.45 and 0.5 m from one of the channel edges.

¹ <https://bluerobotics.com/store/sonars/imaging-sonars/ping360-sonar-r1-rp/>.

² <https://github.com/bluerobotics/ping-protocol>.

Table 2

Experimental configuration of the proposed sonar simulation framework, including the operating specifications of the Ping360 sonar and the main parameters for the omnidirectional screen space reflection algorithm used throughout the validation experiments.

Ping360				SSR			
Parameter	Value	Parameter	Value	Parameter	Value	Parameter	Value
Frequency	750 kHz	Beam width/height	$2^\circ \times 25^\circ$	Ray marching ϵ	0.02	Max. iterations	400
Gain	Medium	Max. range	50 m	Ray step	Adaptive	Initial step size	0.25
Samples per beam	1200	Number of beams	400	Primary buffer	1024×1024	Secondary buffer	512×512

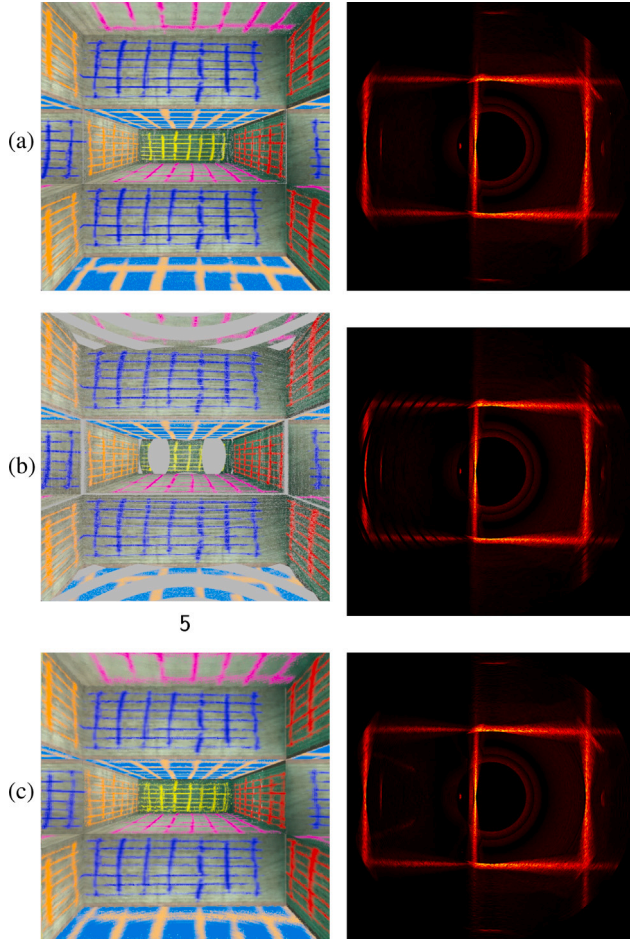


Fig. 4. Visual artifacts produced by different values of the ray-marching tolerance (ϵ) and the initial step size. The test scenario consists of a hollow rectangular enclosure measuring $10 \times 12 \times 8$ m. The left column shows omnidirectional reflection maps used to visualize the simulated secondary reflections, while the right column presents the corresponding 360° sonar scans (400 beams). (a) Reference configuration using the parameters adopted throughout the experiments ($\epsilon = 0.02$, initial step size = 0.25). (b) Effect of increasing the initial step size, leading to missed intersections and ring-shaped artifacts. (c) Effect of increasing the ray-marching tolerance to $\epsilon = 1$, producing degraded reflection quality.

Each of the five scenes proposed for Scenarios A and B represents a complete 360° scan, both for real and simulated sonar insonification, generated by the sequential accumulation of acoustic beams over time. Each scan consists of 400 acoustic beams (one beam per gradian, corresponding to an angular resolution of 0.9°), with each beam containing 1200 range samples. The resulting data are stored as 8-bit images with a resolution of 400×1200 pixels in waterfall format, which can be transformed into a 1200×1200 polar representation for visualization. Real scenarios and their corresponding virtual representations can be seen in Fig. 5.

These scenarios aim to replicate conditions as closely as possible to those of a real-world operational setting. As a result, the vehicle remains active during data collection, with potential sources of noise including the propellers and the resonance of the sonar with the mounting structure and the sonar itself. Additionally, the scenarios include elements such as suspended sediment and reverberations from the hollow metal structures inside the tanks. Furthermore, real-world conditions include small movements during testing, resulting from the ROV's displacement during data acquisition, which have not been compensated for in the simulation. While these sources of noise and unwanted elements are not part of the simulation itself, they are an integral part of the variability of the real environment, allowing for the evaluation of the robustness of the simulation method under real-world conditions.

4.1. Qualitative/quantitative performance

In order to verify the performance of the proposed method, the scenarios have been modeled for use in the NauSim simulator. Within the simulated environment, the Ping360 sonar simulated source has been positioned to ensure that the virtual insonification aligns with the actual insonification recorded in the real-world scenarios.

Fig. 9 shows the results obtained in the various proposed scenarios, both in real and simulated situations, plotted in a polar coordinate system. The official minimum operating range for a Ping360 sonar is estimated at approximately 0.75 m. Within the specified frequency range, transducer “ringing” (internal reflections and sensor resonance) tends to generate a noise zone that can mask returns, making closer measurements unreliable. This zone has been excluded from the tests.

An evaluation is proposed using four different metrics: MSE (Mean Squared Error), PSNR (Peak Signal-to-Noise Ratio), MS-SSIM (Multi Scale Structural Similarity Index), and the distance between HOG (Histogram of Oriented Gradients) descriptors (William T. Freeman, 1994). MSE and PSNR are traditional metrics that measure the fidelity of an image by comparing the absolute mathematical difference pixel by pixel. MSE calculates the average of the squared error (aiming for 0), while PSNR uses a logarithmic scale in decibels to express the ratio between the maximum signal and noise (aiming for high values). Due to the potential for minor discrepancies between virtual and physical scenarios, as well as variations in the sonar's positioning during the insonification, it is recommended to incorporate evaluators that are more resilient to rotations and displacements. Therefore, MS-SSIM is used to evaluate changes in luminance, contrast, and structure between images obtained at different scales of the reference and estimated images. HOG emphasizes the magnitude and direction of the gradient, accentuating edges and corners to define the shape of objects. Geometric similarity is evaluated by measuring the distance (Cosine) between features and HOG descriptors, providing the descriptor values close to one for images with a similar appearance. While the performance of MSE and PSNR is subject to degradation when an image is slightly rotated, scaled or shifted, MS-SSIM and HOG are capable of identifying scenes that are similar to the human eye.

Table 3 shows the performance of the three insonification models in each of the proposed scenarios. It can be observed that the Oren–Nayar model consistently provides the best overall results. This model exhibits the lowest MSE values in all cases and the highest PSNR values, with

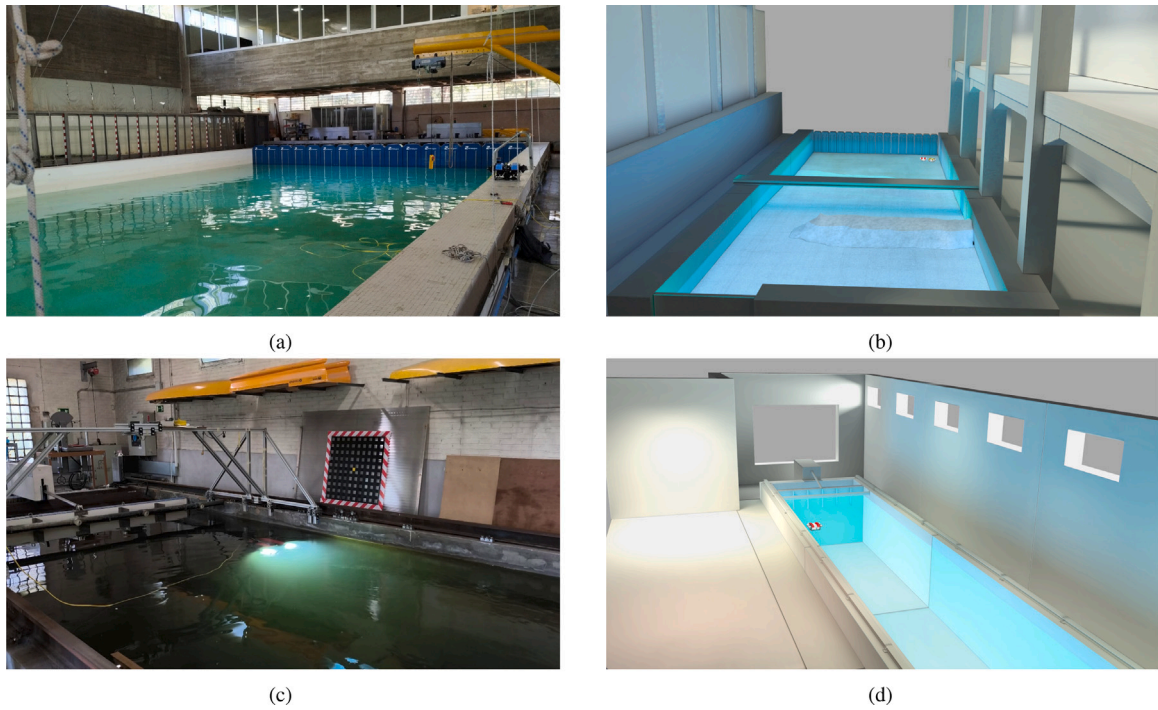


Fig. 5. Real testing environments and their corresponding virtual representations used for validation. (a) shows the Ports Laboratory pool, and (c) the CEHINAV hydrodynamic testing channel, alongside their respective digital twin models (b and d) implemented in the NauSim simulation platform.

Table 3

Quantitative comparison between simulated and real sonar images across all scenarios and three insonification models (Cosine, Squared-Cosine, and Oren-Nayar), using MSE, PSNR, MS-SSIM, and HOG metrics to evaluate both pixel-wise accuracy and structural similarity.

	Cosine				Squared-Cosine				Oren-Nayar			
	MSE↓	PSNR↑	MS-SSIM↑	HOG↑	MSE↓	PSNR↑	MS-SSIM↑	HOG↑	MSE↓	PSNR↑	MS-SSIM↑	HOG↑
Scenario A scene 1	267.5	23.8	0.79	0.79	265.4	23.8	0.8	0.72	234.5	24.4	0.8	0.82
Scenario A scene 2	250.5	24.1	0.78	0.82	278.8	23.6	0.78	0.7	218.3	24.7	0.79	0.84
Scenario A scene 3	248.8	24.1	0.78	0.83	270.6	23.8	0.79	0.7	225.1	24.6	0.78	0.85
Scenario B scene 1	339	22.8	0.71	0.85	377.2	22.3	0.71	0.77	326.3	23	0.72	0.87
Scenario B scene 2	490	21.2	0.74	0.81	580.8	20.5	0.73	0.68	398.5	22.2	0.74	0.84

an average increase of 0.6 dB compared to classic Lambert's cosine law (cosine) and 1 dB compared to squared-cosine. These results suggest a smaller direct discrepancy with respect to the reference. In terms of structural similarity, both MS-SSIM and HOG maintain competitive or superior values, especially in Scenario A, which indicates better preservation of structural fidelity.

The cosine insonification model exhibits intermediate performance. Although in some cases (Scenario A, scenes 1, 2 and 3) it yields results close to Oren-Nayar, it generally exhibits higher MSE values and a slight degradation in MS-SSIM. However, it maintains relatively high values in HOG, especially in the case of Scenario B, indicating good capture of structural fidelity even when the global error is higher. The squared-cosine model generally performs the worst. It exhibits the highest MSE values and the lowest PSNR values, particularly in Scenario B, indicating a reduced ability to model reflectance. Furthermore, the HOG values are consistently lower, indicating a poorer representation of the image's structural features.

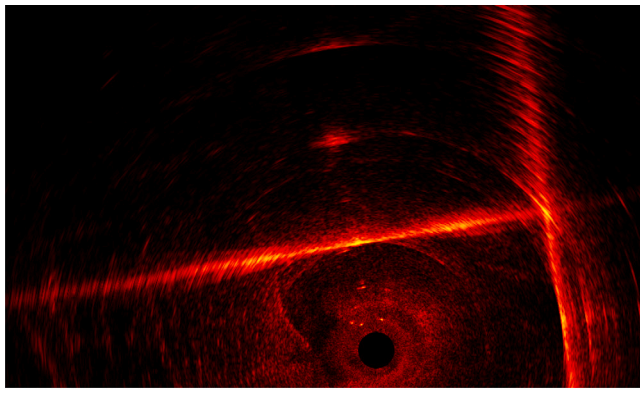
Finally, it should be noted the prevalence of unmodeled noise in the sonar ground truth, particularly in Scenario B. This could explain, to some extent, the high MSE values and the inconsistency of the PSNR across the different environments. Under such circumstances, Oren-Nayar exhibits greater robustness, while squared-cosine experiences significant degradation.

From the perspective of visual analysis, the overall structure of the scenes, including primary and secondary reflections, appears to be correctly captured in all tested methods. In general, there is a tendency

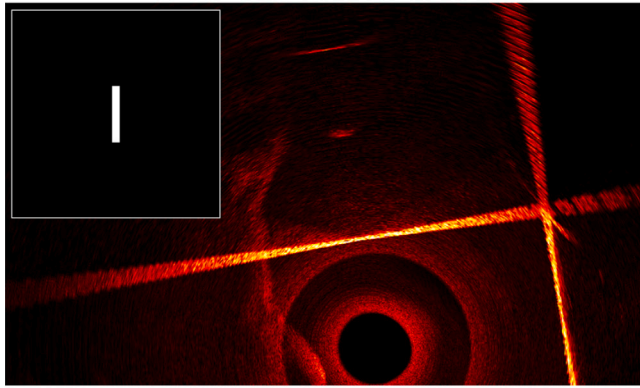
to overemphasize primary reflections relative to ground truth in all three methods, possibly due to the lack of a model for material acoustic absorption. The cosine method shows a certain tendency to overrepresent secondary reflections. (See the results for the cosine method in Scenario B, scenes 1 and 2 in Fig. 9). The use of the squared-cosine method, whose response is significantly affected by variations in the angle of incidence, tends not only to minimize secondary reflections, as can be seen in Fig. 9 Scenario B scenes 1 and 2, but also significantly affects the returns from the background and the air-water interface. (See the results for the squared-cosine method in scenes 1 and 2 for Scenario A in Fig. 9.)

As illustrated in Fig. 6, the presence of side lobes obscures the structures in the image, reduces the intensity of primary and secondary reflections, and alters how the beam interacts with the air-water interface. These interactions are not captured when using only a mask that represents the Ping360's field of view. It should be noted that the beam profile has been derived from a simulation performed in MATLAB, utilizing the characteristics of the Ping360 beam. At the time of writing this paper, no experimental characterization of the emission beam was available.

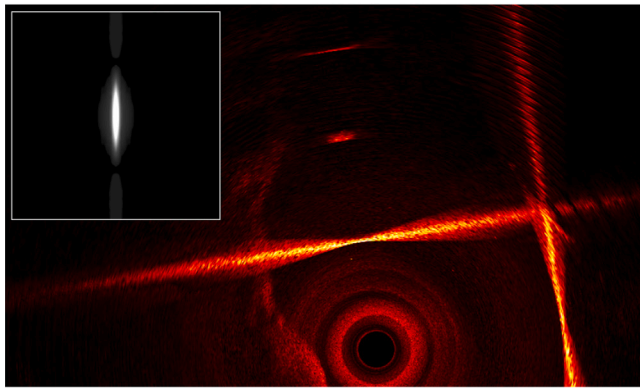
Fig. 7(b) shows the results of the simulation without anti-aliasing, where it can be seen that both primary and secondary reflections tend to form concentric rings. These rings are particularly evident on surfaces with grazing angles of incidence, such as the bottom of the pool. The use of procedural dithering (Fig. 7(c)) and the addition of microtextures in the surface normals (Fig. 7(d)) effectively reduces



(a)



(b)



(c)

Fig. 6. Detail of Scenario A, Scene 2. Comparison between the results obtained in a real-world environment (a) and the simulated images generated using a mask representing only the Ping360 sonar's field of view (25° vertical, 2° horizontal) (b) and a mask representing a model of the beam geometry (c). The masks used are shown in the upper left corner of the corresponding images. Exposure of the mask used in (c) has been adjusted to allow for better visualization.

these artifacts, resulting in a more accurate and detailed representation of acoustic scattering.

4.2. Performance evaluation

To evaluate the computational cost of the simulator, a worst-case scenario is considered, representing the most unfavorable conditions for the execution of the SSR algorithm. In such cases, the algorithm requires long rays (e.g., in a large, open, highly reflective environment),

Table 4

Computational performance of the proposed sonar simulation method under different primary and secondary buffer resolutions, reported in terms of frame time and frame rate.

buffer1/buffer2	RTX 2000 Ada		RTX 3060	
	ms/frame	Hz	ms/frame	Hz
256 × 256/256 × 256	9	110	15	66
256 × 256/512 × 512	19	52	29	35
256 × 256/1024 × 1024	35	29	50	20
512 × 512/256 × 256	11	92	18	51
512 × 512/512 × 512	20	50	30	33
512 × 512/1024 × 1024	36	28	54	18
1024 × 1024 /256 × 256	14	72	23	42
1024 × 1024/512 × 512	21	41	34	29
1024 × 1024/1024 × 1024	40	25	59	17

increasing the number of steps needed to trace each ray and, consequently, the computational cost (Hofmann et al., 2017). This scenario serves as the basis for evaluating the algorithm's performance.

The simulation is configured with a range of 50 m, corresponding to the theoretical maximum range of the Ping360 sonar, within a large open environment.

Under these conditions, performance is primarily determined by the number of pixels used to compute both primary and secondary reflections. This, in turn, depends on the resolution of the buffers associated with the main and secondary cameras. Multiple buffer configurations are evaluated. Considering the insonification mask applied, a buffer of 256 × 256 corresponds to 6311 pixels, a 512 × 512 buffer corresponds to 24,184 pixels, and a 1024 × 1024 buffer corresponds to 91,548 pixels.

The tests are conducted on two different systems: a laptop equipped with an Intel i7-13850HX (2.10 GHz), 32 GB RAM, and an NVIDIA RTX 2000 Ada Generation GPU, and another laptop with an Intel i7-12700H (2.70 GHz), 32 GB RAM, and an NVIDIA RTX 3060 GPU. The evaluation metrics used are frame rate and average frame time.

The results are presented in Table 4, where each test corresponds to the average performance over 400 rays, representing a full 360-degree scan around the sonar.

These results show that computational cost increases with buffer resolution, leading to higher frame times and lower frame rates. The resolution of the secondary buffer has a stronger impact on performance than that of the primary buffer. For instance, when the primary buffer is fixed at 256 × 256, increasing the secondary buffer from 256 × 256 to 1024 × 1024 raises the frame time from 9 ms to 35 ms in the RTX 2000 Ada configuration, and from 15 ms to 50 ms in the RTX 3060 configuration. This behavior is consistent across all buffer configurations, indicating that secondary reflections are the dominant factor in computational cost, as they require additional ray-marching steps.

Real-time performance (above 30 Hz) is maintained in most configurations, except when the secondary or both buffers are set to high resolutions, reaching the worst-case scenario at 1024 × 1024. However, it is important to note that the refresh rate of a sonar system depends on its hardware specifications and the configuration of the insonification process. Additionally, as expected, the RTX 2000 Ada configuration consistently outperforms the RTX 3060 configuration, with larger differences under higher computational loads, indicating better scalability of the more advanced GPU.

5. Discussion and conclusions

This paper presents a novel approach to real-time sonar simulation based on rasterization and an omnidirectional implementation of screen space reflections (SSR), integrated into the NauSim simulation platform. The proposed methodology reinterprets acoustic propagation

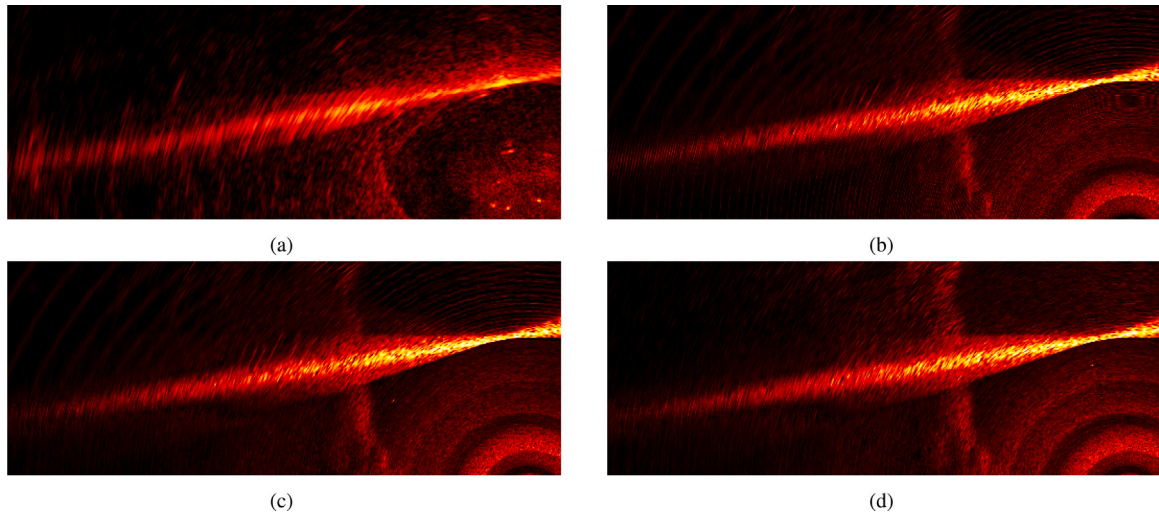


Fig. 7. Mitigation of depth-buffer aliasing and banding. Detail of Scenario A, scene 2. Comparison of the results obtained in a real environment (a) with the simulated image without any anti-aliasing (b), the simulated image with procedural dithering applied to the primary reflections (c), and the simulated image with procedural dithering applied to primary reflections and microtextures on the surface normals (d).

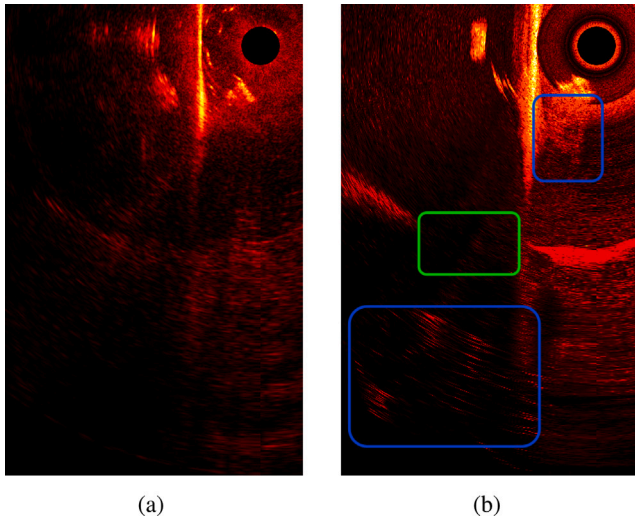


Fig. 8. Detail of Scenario A, scene 3. Inherent limitations of the screen space reflections method. Comparison between the results obtained in a real-world environment (a) and the simulated image (b). The blue-highlighted regions show noise resulting from a lack of information about the geometry of the back faces and/or the low resolution of the ray-casting. The green-highlighted areas display missing reflection data or “holes” caused by occluded geometry, which are not captured in the underlying depth buffers. The exposure in image (b) has been increased to improve visibility.

as a reflection problem in image space, allowing primary and secondary echoes to be generated at a significantly reduced computational cost compared to traditional ray-tracing techniques. It also includes a revision of the classic Lambertian sound-scattering model, an approximation of the presence of side lobes in the signal, and solutions to mitigate aliasing and banding artifacts. These enhancements facilitate the simulator’s ability to reproduce complex sonar phenomena frequently observed in real underwater environments.

The experimental validation process was conducted using data obtained with a Ping360 sonar mounted on a BlueROV2 platform. Absolute error metrics, such as MSE and PSNR show sensitivity to slight misalignments and unmodeled environmental noise. However, comparisons using structural similarity metrics (MS-SSIM and HOG descriptor) indicate that the simulated scenes retain the relevant geometric features

and echo structures observed in actual measurements. In particular, the Oren-Nayar insonification model produced more stable and accurate results compared to general Lambertian and squared-cosine models.

The goal of this work is not to accurately simulate every aspect of underwater acoustic propagation, but to achieve a practical balance between computational efficiency and acoustic realism by generating sonar imagery that preserves the structural characteristics and dominant propagation effects relevant to perception, navigation, and algorithm development. Consequently, the proposed framework should be interpreted as a sonar image simulator based on an approximation of general acoustic signal behavior, rather than as a rigorous physical simulation of underwater acoustics. As with any image-space approximation, the method inherits the limitations associated with this level of abstraction.

First, acoustic propagation is approximated rather than physically modeled. The proposed framework does not solve the acoustic wave equation, nor does it explicitly account for wave interference or frequency-dependent propagation effects. While the Oren-Nayar insonification model improves upon purely Lambertian formulations by more accurately representing scattering from rough surfaces, it remains a simplified reflectance model and is unable to fully reproduce the complex mechanisms involved in underwater acoustic backscattering. Although non-modeled environmental noise sources, such as vehicle structural resonance, propeller vibration, and test-tank reverberation, must be considered when interpreting the experimental results, the unsatisfactory MSE and PSNR values should not be overlooked, as they indicate that relevant aspects of acoustic image formation are not being fully captured. This highlights the need for future developments focused on improving the physical realism of the acoustic model.

Furthermore, several environmental phenomena affecting acoustic images have been omitted in favor of computational efficiency. The current implementation does not model water-column reverberation, suspended-particle scattering, thermocline-induced refraction, or other volumetric propagation effects. In particular, acoustic attenuation and absorption are represented by a simplified approximation rather than a physically based propagation formulation, which limits the simulation’s accuracy.

These limitations suggest some promising areas to focus on in future research. Although a full physical simulation of underwater acoustics is beyond the scope of this work, the proposed framework could be expanded to include more sophisticated propagation models while maintaining the efficiency of the graphics pipeline. For example, image-space cone marching or cone tracing techniques could efficiently

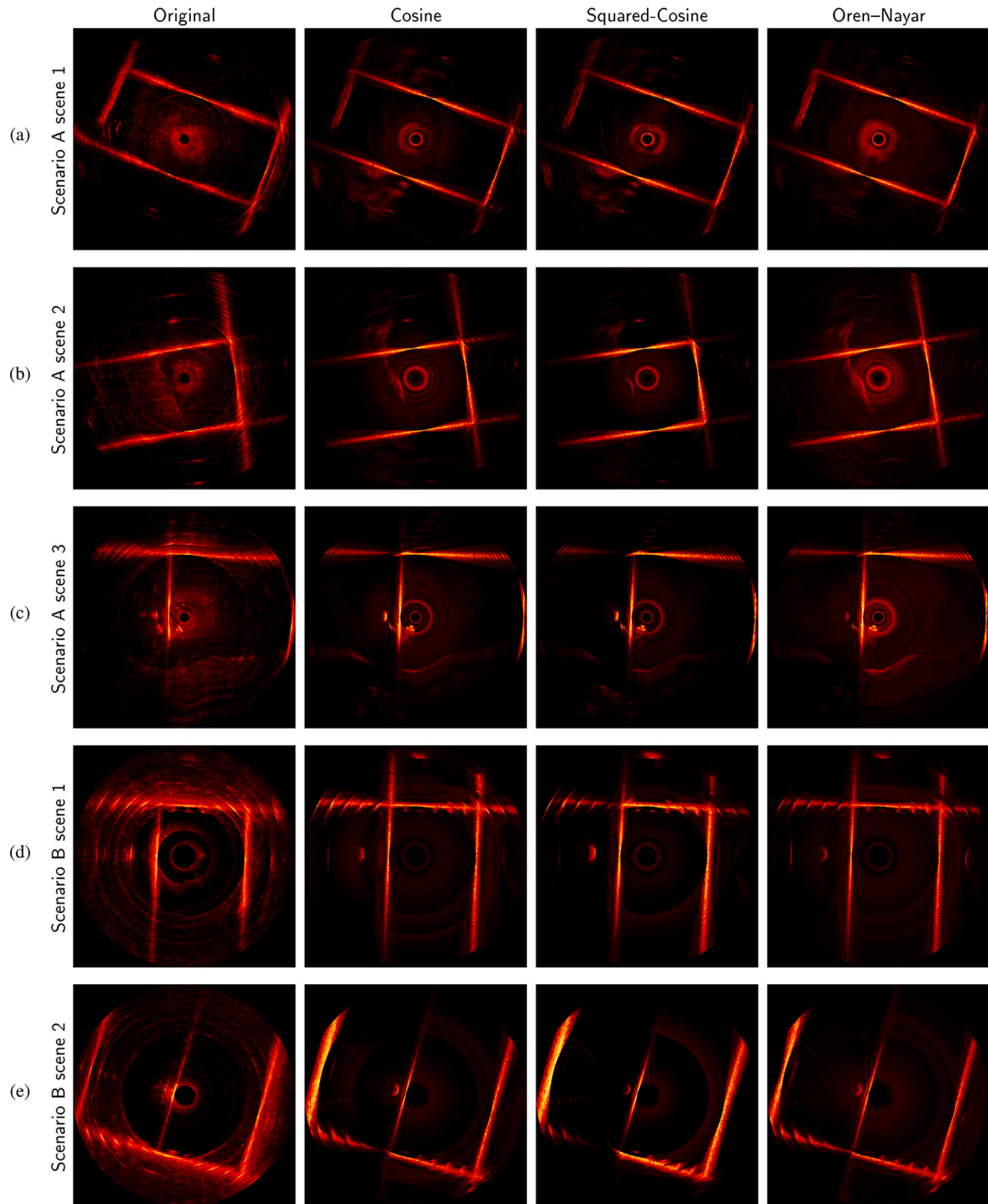


Fig. 9. Qualitative comparison of real and synthetic sonar images in the proposed test scenarios. The columns show the results obtained in the real-world test environments and those of the three insonification models tested: Cosine, Squared-Cosine, and Oren-Nayar.

approximate volumetric acoustic propagation by accounting for beam divergence and accumulated scattering. Furthermore, more advanced scattering models could complement or replace the Oren-Nayar formulation to more accurately reproduce the angular dependence of acoustic backscatter from various seabed materials and man-made structures. Similarly, volumetric ray marching approaches (Beyer et al., 2015) could be employed to simulate suspended sediments, water-column reverberation and other environmental effects that significantly influence sonar images in natural scenarios.

Another limitation concerns experimental validation. The proposed framework has been evaluated using a single mechanically scanning

imaging sonar (Ping360) in two controlled hydrodynamic testing facilities. While these environments provide adequate acoustic conditions for direct comparison between measured and simulated sonar images, their environmental diversity remains limited. Further validation in open-water scenarios, including natural seabeds, cluttered harbor environments and offshore infrastructure, would be valuable in assessing the generalization capability of the proposed approach.

It is also worth noting that the dominant sources of acoustic degradation differ between controlled environments and open water. In testing tanks, multipath reverberation caused by surrounding structures is often the main cause of image degradation. In contrast, open-water

scenarios are more strongly influenced by attenuation, volume scattering and environmental variability. The shader-based architecture proposed in this work provides a suitable basis for incorporating these effects as additional rendering stages, without necessitating fundamental modifications to the simulation pipeline.

Finally, it is important to highlight the limitations associated with the use of SSR. Despite extending the system to cover the entire 360° environment, SSR remains limited to the information available in the depth/frame buffers. Elements occluded by other objects or reflections from the rear of an object remain imperceptible to the method, which can result in omissions and artifacts, as shown in Fig. 8. While there are ways to mitigate these issues (Beug, 2020), they often involve a significant performance cost. Possible alternatives include using parallax-corrected cubemaps (Sébastien and Zanuttini, 2012) as a basis for calculating reflections or transitioning toward hardware-accelerated ray tracing on RTX platforms.

Future work will therefore focus on improving the physical fidelity, environmental robustness and applicability of the proposed framework. In addition to incorporating more realistic acoustic propagation and scattering mechanisms, the model will be adapted for use with other sonar technologies, including side-scan sonar and multibeam echosounders. This implies extending the pipeline to account for modality-specific characteristics such as beam geometry, multibeam crosstalk and multipath interactions.

In conclusion, this work presents a modular, efficient, and scalable framework for generating synthetic sonar images for mechanically scanning imaging sonar systems. The implementation demonstrates the potential of modern graphics pipelines to approximate relevant acoustic phenomena, including backscattering, side lobes, and multipath propagation, while reducing common rendering artifacts such as aliasing. The results validate the framework as a useful tool for training and evaluating object perception and detection algorithms, where the preservation of target geometry, edges, and acoustic shadows may be more relevant than exact pixel-level intensity agreement.

CRediT authorship contribution statement

César Antonio Ortiz-Toro: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rodrigo Saldaña-Esteban:** Conceptualization, Investigation, Validation, Resources. **Juan Parras:** Writing – review & editing, Supervision, Project administration. **Álvaro Gutiérrez:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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